Exam Date & Time: 26-Sep-2023 (02:45 PM - 04:45 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

Mid-Term Examination Sep 2023// Engineering Mathematics III (MAT2122)

ENGINEERING MATHEMATICS - III [MAT 2122]

Marks: 30 **Duration: 120 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

Answer all the questions

1) The constant term in the Fourier series expansion of the function $f(x)=\sin\left(\frac{x}{2}\right)$ in the interval $0\le x\le 2\pi$ is (0.5)

1)	$\frac{2}{\pi}$		2)	$\frac{-2}{\pi}$		3)	$\frac{1}{\pi}$		4)	$\frac{-1}{\pi}$	

2) The primitive period of the periodic function $f(x)=\sin 6x+\cos 12x$ is (0.5)

1)	$\frac{\pi}{3}$		2)	$\frac{\pi}{6}$		3)	π		4)	$\frac{\pi}{18}$	

3) The value of the Fourier series of the function $f(x)=\begin{cases}-\pi & if\ -\pi<x\le 0\\ x & if\ 0<x<\pi\end{cases}$ at x=0 is (0.5)

1)	$\frac{-\pi}{2}$		2)	π		3)	$-\pi$		4)	$\frac{\pi}{2}$	

4) If A is a square matrix of order 2 such that $trace(A)=2$ and $determinant(A)=-3$, then (0.5)

1)	3 and -1 are the eigenvalues of A.		2)	-3 and 1 are the eigenvalues of A.		3)	1 and -2 are the eigenvalues of A.		4)	-1 and 2 are the eigenvalues of A.	

5) The half range Fourier sine series of the function $f(x)=\begin{cases}x & if\ 0<x\le 1\\ 1 & if\ 1<x<2\end{cases}$ at x=0 converges to (0.5)

1) 0	2) 1	3) 0.5	4) 1.5

6) If $F_s(e^{-x}) = \sqrt{\frac{2}{\pi}}\left(\frac{s}{1+s^2}\right)$, then $F_c(xe^{-x})$ is (0.5)

1) $\sqrt{\frac{2}{\pi}}\left(\frac{2s}{(1+s^2)^2}\right)$	2) $\sqrt{\frac{2}{\pi}}\left(\frac{-2s}{(1+s^2)^2}\right)$	3) $\sqrt{\frac{2}{\pi}}\left(\frac{1-s^2}{(1+s^2)^2}\right)$	4) $\sqrt{\frac{2}{\pi}}\left(\frac{s^2-1}{(1+s^2)^2}\right)$

7) If $F\{f(x)\} = F(s)$, then $F\{f(-2x)\}$ is. (0.5)

1) $\frac{1}{2}F\left(\frac{-s}{2}\right)$	2) $\frac{1}{2}F\left(\frac{s}{2}\right)$	3) $-F(2s)$	4) $2F(-2s)$

8) The constant term in the Fourier series expansion of the function y=y(x), $0 \le x \le 6$, computed using the following data is (0.5)

x	0	1	2	3	4	5
y	14	11	5	16	8	2

1) 9.3333	2) 11.2	3) 8	4) 14

9) Which one of the following matrices is NOT diagonalizable? (0.5)

1) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$	2) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$	3) $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$	4) $\begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix}$

10) Which one of the following statements is CORRECT? (0.5)

1) The matrices $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ are similar.	2) The matrices $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ are similar.	3) The matrices $\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ are similar.	4) The matrices $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are similar.

B

Answer all the questions.

Answer all the questions

11) Find the Fourier cosine and sine transform of $f(x) = \frac{1}{\sqrt{x}}$, $0 < x < \infty$. (3)

12) (3)

Find the Fourier transform of $f(x)=\begin{cases}1-x^2, & |x|\le 1\\ 0, & |x|>1\end{cases}$ and hence evaluate

$$\int_0^{\infty}\frac{\sin t - t\cos t}{t^3}dt.$$

13) Find the Fourier series expansion of

$f(x)=\begin{cases}x, & 0<x<\pi\\ 2\pi-x, & \pi<x<2\pi\end{cases}$, $f(x+2\pi)=f(x)$ and hence deduce that $1+\frac{1}{3^2}+\frac{1}{5^2}+\ldots=\frac{\pi^2}{8}$. (3)

14) Find the Cholesky decomposition of A=$\begin{bmatrix}4 & 10 & 8\\ 10 & 26 & 26\\ 8 & 26 & 61\end{bmatrix}$. (2)

15) The displacement y of a part of a mechanism is tabulated with corresponding angular movement x degrees of a crank. Express y as Fourier series upto second harmonics. (4)

x (in degrees)	0	60	120	180	240	300
y	1.8	0.3	1.5	2.1	1.3	1.76

16) Find an orthogonal matrix P such that $A=PDP^T$, where D is a diagonal matrix and

$A=\begin{bmatrix}8 & -2 & 2\\ -2 & 5 & 4\\ 2 & 4 & 5\end{bmatrix}$. (4)

17) Express f(x)=x as a half range Fourier cosine series in 0< x< 2. (3)

18) Diagonalize the matrix $A=\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}$ and hence find $\sqrt{A}$. (3)

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